

CAS: Non-parametric Morphology measurements

CAS - System

C: Concentration $C \equiv 5 \log \left(\frac{r_{80}}{r_{20}} \right)$
 r_{80} : radius that contains 80% of total light
 r_{20} : " " 20% "

• Tells us how much light is found in bulk

ellipticals: $4 < C < 5$

Spirals: $3 < C < 4$

A: (A) - Symmetry

Rotate image 180° and subtract from original

$$A \equiv \frac{|I - R|}{I}$$

$$0 < A < 1$$

I: original image

R: Rotated image

$A = 0$: totally symmetric

$A = 1$: totally Anti-symmetric

S: Subtraction

Subtracting original image from a smoothed or blurred version

$$S \equiv \frac{I - B}{I}$$

$$0 \leq S \leq 1$$

$S = 0$: Smooth

$S = 1$: clumpy

Dynamics:

(i) Virial Theorem (VT)

$$2K + U = 0$$

$$\bullet K = \frac{1}{2} \sum m_i v_i^2 = \frac{1}{2} M_{gal} \langle v^2 \rangle$$
$$\langle v^2 \rangle = \langle v_x^2 + v_y^2 + v_z^2 \rangle = 3 \langle v_r^2 \rangle = 3 \langle \sigma_r^2 \rangle$$

$$\underline{K} = \frac{1}{2} M_{gal} \langle v^2 \rangle = \underline{\frac{3}{2} M_{gal} \sigma_r^2}$$

• General potential Energy

$$U(R) = -\alpha \frac{G M_{gal}^2}{R}$$

• From Vir. Thm:

$$2 \left(\frac{3}{2} M_{gal} \langle v^2 \rangle \right) = \alpha \frac{G M_{gal}^2}{R}$$

$$\Rightarrow M_{gal} = \frac{3R \langle v^2 \rangle}{\alpha G} = \hat{c} R \langle v^2 \rangle$$

Constant

* This is true in general. So, let's look at spirals and then ellipticals:

Spirals:

(i) Has $I(R) = I_0 e^{-R/R_h}$

$$L_{max} = \int_0^{2\pi} d\phi \int_0^{\infty} R e^{-R/R_h} dR$$

L_{max} : is the maximum possible output so we take the integral to $\rightarrow \infty$

$$L_{max} = 2\pi I_0 R_h^2$$

→ Assumption: We assume all spirals have similar profiles and are close to $L_{\max}$

$$L = C_1 I_0 R^2$$

C_1 : some constant



$$\Rightarrow R = \sqrt{\frac{L}{C_1 I_0}}$$

From VT: $M_{gal} = \tilde{C} R \langle V^2 \rangle$

→ Assumption: $\langle V^2 \rangle \sim V_{\max}^2$ measured and determined by HI

so $M_{gal} = \tilde{C} R V_{\max}^2$

• We want to know the mass to light ratio

$$\frac{M}{L} \equiv \mu$$

so divide above by L :

$$\frac{M_{gal}}{L} = \tilde{C} \frac{R V_{\max}^2}{L}$$

$$\mu = \tilde{C} \frac{R V^2}{L}$$

$$L = \frac{\tilde{C} R V^2}{\mu}$$

Now, from  we have $R = \frac{L^{1/2}}{C_1^{1/2} I_0^{1/2}}$ so

plug this in

$$L = \frac{\tilde{C}}{\mu} \left(\frac{L^{1/2}}{C_1^{1/2} I_0^{1/2}} \right) V^2$$

new constant:

$$C_2 \equiv \frac{\tilde{C}}{C_1^{1/2}}$$

$$L^{3/2} = \frac{C_2}{\mu} \frac{V^2}{I_0^{1/2}}$$

square both sides:

$$L = C_3 \left(\frac{V^4}{M^2 F_0} \right)$$

(Tully-Fisher)

$$C_3 = C_2^2$$

- Assumptions: F_0 : constant
 M : constant

Ellipticals:

- Similar analysis

$$L_{\max} = (7.22) I_c R e^2$$

→ All the same assumptions

- From Vir. Theorem

$$M_{gal} = \tilde{c} R \sigma_r^2$$

here we have the dispersion σ_r^2 rather than V_{max} from spirals

- divide by L :

$$\frac{M_{gal}}{L} = \tilde{c} \frac{R \sigma_r^2}{L}$$

$$\mu = \tilde{c} \frac{R \sigma_r^2}{L}$$

solve for L : $L = \frac{\tilde{c} R \sigma_r^2}{\mu}$

- Assumption: μ is a constant, so

$$\frac{\tilde{c}}{\mu} = C_2$$

so

$$L = C_2 R \sigma_r^2$$

• Assumption: All ellipticals have a similar profile so

$$L = C_1 I_e R^2$$

Equating these two

$$C_2 R \sigma_r^2 = C_1 I_e R^2$$

let

$$C_3 \equiv \frac{C_2}{C_1}$$

$$C_3 \sigma_r^2 = I_e R^2$$

• Take $\ln$:

$$\ln(C_3) + 2 \ln(\sigma_r) = \ln(I_e) + \ln(R)$$

* Now, R_e is where half of the light of M_{gal} is contained and so we often let $R = R_e$:

$$\ln(C_3) + 2 \ln(\sigma_r) = \ln(I_e) + \ln(R_e)$$

solve for R_e :

$$\ln(R_e) = 2 \ln(\sigma_r) - \ln(I_e) + C_4$$

(Faber-Jackson)

↑
 $C_4 = \ln(C_3)$

